xAAD in Practice: Explainable Anomaly Detection in Cybersecurity

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***Abstract*—In this paper, we demonstrate the practical deployment of the Explainable Active Anomaly Discovery framework (xAAD) in a live cybersecurity setting, highlighting how its explainability and active feedback capabilities directly reduce false positives, streamline threat triage, and enhance the overall trust and efficiency of security operations centers (SOCs). By evaluating xAAD on the RoEduNet-SIMARGL2021 dataset, we illustrate its immediate operational benefits and provide guidance for its integration into existing cybersecurity workflows. Our findings indicate that xAAD's explanations are both sparse—focusing on a minimal set of influential features—and robust to data perturbations, offering stability across dynamic network conditions. These qualities aid security analysts in making faster, more informed decisions, ultimately reducing investigation overhead. By bridging technical anomaly detection outputs with the operational needs of security operations centers, xAAD represents a valuable step toward practical, explainable anomaly detection solutions in high-stakes cybersecurity environments.**

Keywords*—anomaly detection, explainability, isolation forest, active anomaly discovery, cybersecurity

I. Introduction

Anomaly detection has become integral to modern cybersecurity [1] [2] [3], as it can uncover malicious activities that bypass traditional signature-based defenses. Yet many existing algorithms, including widely used methods like Isolation Forest [4], produce opaque outputs, leading to high false positive rates and prolonged investigations. In a typical Security Operations Center (SOC), such opacity results in alarm fatigue and inefficient resource allocation. By offering clear, feature-level explanations, xAAD empowers analysts to quickly discern the root causes of anomalies, thereby enhancing response speed and overall trust in the system.

Active Anomaly Discovery (AAD) [5] [6] [7] addresses some of these concerns by incorporating domain-expert feedback to iteratively refine anomaly scores. However, standard AAD solutions typically do not generate feature-level explanations, making it difficult to pinpoint precisely which network behaviors or attributes triggered an alert. While our previous work detailed the algorithmic design and theoretical benefits of xAAD [8], this paper focuses on its practical deployment in cybersecurity. We demonstrate how xAAD's post-feedback explainability enhances operational decision-making, reducing false positives and offering clear insights that are directly actionable for security analysts.

Here, we deploy xAAD on the RoEduNet-SIMARGL2021 intrusion detection dataset and evaluate its operational impact using three key E-XAI metrics—Descriptive Accuracy, Sparsity, and Robustness [9]. Our results demonstrate that xAAD not only improves detection performance but also delivers clear, actionable explanations that help security analysts rapidly focus on critical network features, thereby reducing false positives and streamlining threat triage.

II. Related Work

Prior research on explainable artificial intelligence (XAI) has largely focused on generating either local or global explanations for model decisions. Methods such as LIME [10] and SHAP [11] have been widely adopted to interpret predictions of complex black-box models across various domains, including cybersecurity. In the context of cybersecurity, however, explainable systems that incorporate active learning components remain relatively underexplored.

Recent advances include the development of evaluation frameworks specifically tailored for XAI in cybersecurity, such as the work in [9], which introduces metrics like Descriptive Accuracy, Sparsity, Stability, Robustness, Efficiency, and Completeness. Moreover, approaches incorporating analyst feedback, notably the Active Anomaly Discovery (AAD) framework [12], have demonstrated that iterative expert feedback can refine anomaly detection. Building on these ideas, our previous work on the xAAD approach [8] integrates post-feedback metrics into an Isolation Forest-based detector to provide targeted and reliable explanations.

A. Overview of xAAD Framework

The xAAD framework combines two core ideas: (1) Active Anomaly Discovery (AAD), which iteratively leverages expert feedback to refine anomaly scores, and (2) a post-feedback explainability component that highlights the specific features most responsible for each anomaly flag. An anomaly detector (e.g., an Isolation Forest) ranks instances by their anomalousness in a typical AAD setup. A small fraction of highly ranked instances are then shown to domain experts (or a simulated oracle), who label them as true or false anomalies. These labels are used to reweight the underlying ensemble of decision trees, thereby refining the anomaly scores and ideally boosting precision in the

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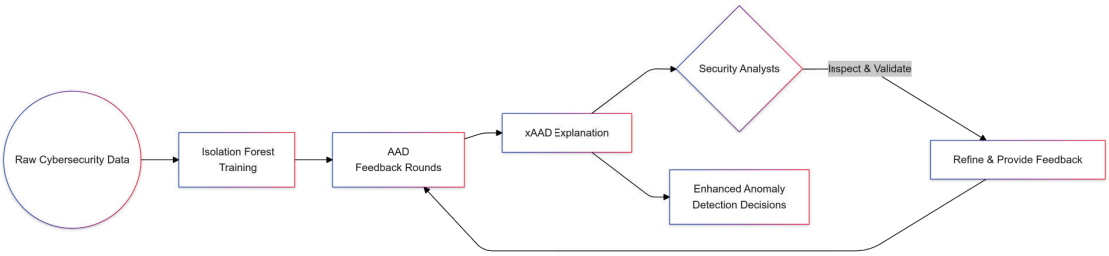

Fig. 1. Flow diagram illustrating the xAAD pipeline applied to the SIMARGL2021 dataset.

next round of feedback. However, classic AAD solutions often stop short of explaining why certain points are flagged.

xAAD builds on our previous work [8] by adding a local explainability module that computes per-feature importances on top of the newly reweighted Isolation Forest. Our prior research demonstrated how even a simple additive term in the per-node weighting scheme can reflect domain feedback; xAAD extends this by directly outputting an explanation vector for each flagged sample. This vector indicates how each feature pushes the sample toward being anomalous (positive contribution) or normal (negative contribution). The key benefit is that these explanations incorporate the new knowledge gleaned from expert labels. This allows security analysts to trace anomalies back to specific packet-level or session-level features—such as unusual port usage or flow duration—rather than receiving a single numeric score.

Despite these advances, systematic evaluations of XAI methods in real-world cybersecurity settings remain limited. In this work, we bridge this gap by not only assessing xAAD's performance using established E-XAI metrics but also by comparing it against baseline models, thereby offering deeper insights into the trade-offs between detection performance and explainability.

III. Dataset and Experimental Setup

We chose the RoEduNet-SIMARGL2021 Network Intrusion Detection Dataset [13] because it reflects the complexities of an active academic network, encompassing both benign traffic patterns and realistically staged cyberattacks. To ensure unbiased model evaluation and robust generalization, we first randomly divided the full dataset into training, validation, and testing subsets with an 80/10/10 split. Unlike fully synthetic datasets, which can fail to capture the unpredictability of authentic environments, RoEduNet-SIMARGL2021 offers genuine network behavior across a large time window, making it highly suitable for evaluating both the performance and practicality of anomaly detection algorithms in real-world scenarios.

To manage the computational demands of our experiments while preserving the dataset's natural variability, we extracted a random subset of 10,000 rows. This sample was then meticulously cleaned to ensure the integrity of subsequent analyses. First, we removed rows containing missing or infinite values to avoid potential biases or numerical instabilities that might arise during model training. A random shuffle was applied afterward so that the order of instances would not inadvertently influence the learning process. We subsequently standardized all numeric features to place them on a comparable scale, which is particularly important for algorithms like Isolation Forest that assume features contribute equally when measuring outlier tendencies. We also pruned out highly correlated features to reduce redundancy and prevent certain dimensions from exerting disproportionate influence on the detection model. These steps collectively help improve robustness, facilitate more interpretable outcomes, and ensure that the final subsets of data are well-suited to both baseline anomaly detection procedures and the iterative, feedback-driven nature of Active Anomaly Discovery (AAD).

IV. Methodology

To better understand the relationships among features in the cleaned 10k subset, we first computed the correlation matrix, shown in Figure 2. This matrix confirms that while certain features retain some degree of correlation, those with near-duplicate information were removed in preprocessing. Reducing redundancy at this stage helps avoid skewed results when training anomaly detection models.

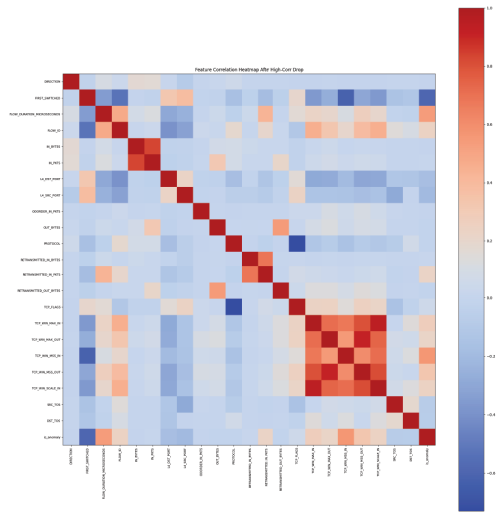

Fig. 2. Correlation matrix of the SIMARGL2021 10k sample after scaling and removing highly correlated features.

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Subsequently, we employed t-SNE to visualize the data distribution in a two-dimensional space. This step allowed us to see whether suspicious traffic patterns would form distinct clusters, offering an intuitive snapshot of how the dataset might separate into normal and abnormal regions. We then tested three different approaches to anomaly detection and explanation. In the first, we simply projected the raw data via t-SNE without training any model, relying on no labels to guide the visualization. In the second, we used a custom Isolation Forest before any active feedback, establishing a baseline detector. Finally, in the third approach, we updated the Isolation Forest with iterative analyst feedback through Active Anomaly Discovery (AAD) to obtain xAAD, which provides local explanations for each anomalous instance it flags.

Figure 1 depicts the main steps of our approach. In the initial Isolation Forest Training phase, we train an ensemble of Isolation Trees on the *train* portion of the data. This creates a rough separation between normal and potentially anomalous points based on how easily individual observations can be isolated. We then initiate a series of AAD Feedback Rounds. In each round, the top-ranked anomalies are presented to a domain expert (or, in our experiments, a simulated oracle that mimics expert judgment) who labels each instance as truly anomalous or nominal. This feedback is used to reweight the ensemble—emphasizing features consistently associated with true anomalies—thus progressively refining the anomaly scores and enhancing the interpretability of the resulting explanations. Once this iterative process converges, we arrive at xAAD Explainability, in which per-instance feature importances can be generated. These importances help clarify *why* a particular connection or flow was flagged as anomalous, aiding human analysts in prioritizing and investigating alerts.

Fig. 3. Descriptive Accuracy: F1 scores before and after xAAD feedback, when top-k explained features are excluded.

A. Evaluation Metrics

Our evaluation draws on three core metrics from [9] to assess both the performance and clarity of xAAD's explanations. Descriptive Accuracy measures whether the top features flagged by xAAD are truly influential in driving anomaly detection. Practically, we remove the most important features and observe how drastically the detection performance (in terms of F1 score) declines. A sharp drop implies that these features are indeed critical to the model. Sparsity checks whether the model relies on only a small fraction of features to explain most anomalous events. This behavior is desirable because it helps security analysts quickly pinpoint which attributes to investigate.

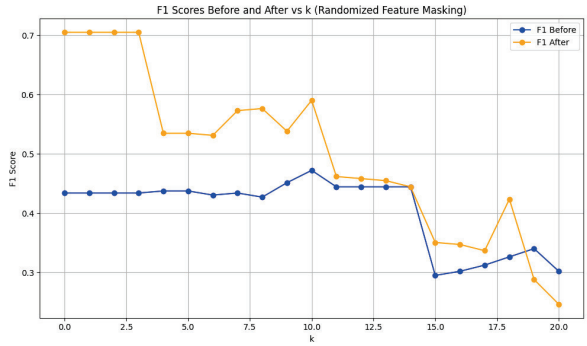

Fig. 4. Comparison: F1 scores when *random* features are excluded (instead of top-xAAD features). Note the milder decline in performance.

Lastly, Robustness involves introducing controlled perturbations (e.g., adding noise) to the data and seeing how stable the resulting feature importances remain. If explanations significantly change under minor alterations, they may be less reliable for real-world cybersecurity applications. Together, these three metrics offer a well-rounded perspective on how effectively and reliably xAAD explains its detection decisions.

V. Results and Discussion

A. Visualization and Anomaly Detection

Figure 5 shows side-by-side t-SNE projections of the processed 10k subset in three different stages. In the first view (Figure 5 - No Training), the raw dataset is plotted without any anomaly labels, providing a baseline visualization of inherent data structure. The second view (Figure 5 - Custom Isolation Forest) depicts anomaly labels generated by a custom Isolation Forest trained on the *train* subset but *before* any analyst feedback. Here, pockets of anomalous points appear, though they may still be interspersed with normal points due to the model's initial uncertainty.

The final view (Figure 5 - xAAD) illustrates data after applying xAAD, in which the Isolation Forest has been iteratively refined via AAD feedback. Visually, the anomalous points are now more cleanly separated from normal points, suggesting that expert corrections guided the model toward a more discriminative boundary. This improved separation is critical in high-stakes scenarios like cybersecurity, where a tighter clustering of anomalies can help reduce false positives and focus analysts' attention on truly suspicious events.

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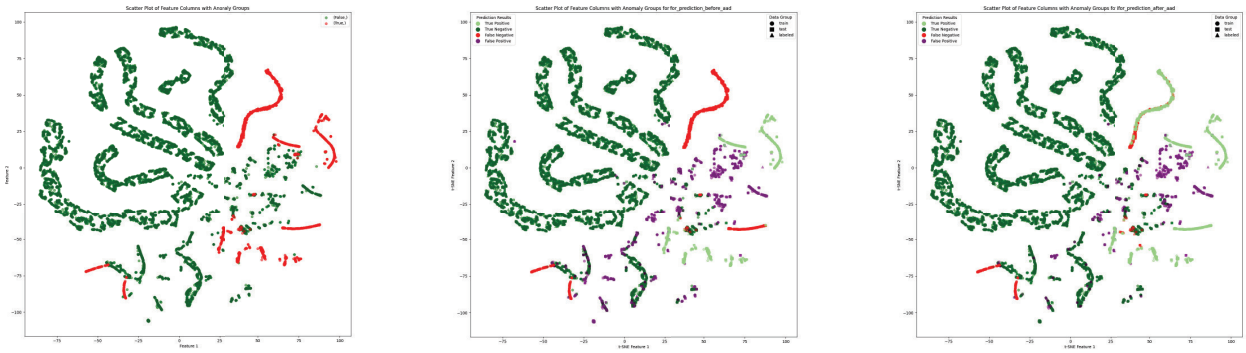

No Training (Raw Data) Custom Isolation Forest (Pre-AAD) xAAD (Post-AAD)

Fig. 5. Side-by-side comparison of t-SNE projections for the SIMARGL2021 10k sample under three different configurations. In (a), the raw dataset is projected without anomaly labels. In (b), points are labeled by a custom Isolation Forest model before Active Anomaly Discovery (AAD). In (c), the Isolation Forest is updated with analyst feedback (xAAD), leading to improved separation of anomalous points. True negatives are colored green, true positives in light blue, false negatives in red, and false positives in purple.

B. Descriptive Accuracy (Metric 1)

To assess the *Descriptive Accuracy* of xAAD's feature-level explanations, we removed the top-k features (as indicated by xAAD importances) and measured how much the F1 score changed. Figures 3 and 4 show the performance drop when the most important features versus random features are excluded, respectively.

The sharper decline observed when removing xAAD's top features demonstrates that these features are indeed crucial for the model's anomaly-detection ability. Conversely, excluding a random set of features leads to a more gradual degradation. In a real-world cybersecurity setting, knowing which features carry the greatest explanatory power allows security teams to prioritize resources effectively. For instance, if the model consistently deems certain network-flow attributes (e.g., byte counts or protocol usage) as most indicative of an attack, analysts can scrutinize logs and firewall configurations related to those attributes first, accelerating incident-response workflows.

Finally, we quantitatively compare the F1 gains in both cases by analyzing the areas under the curve (AUC) for the ΔF1 plots. The AUC for the sorted ranking is 0.6840, while for the random ranking it is 1.9201. The ratio of sorted to random AUCs is 0.3562, demonstrating that xAAD's top features have a substantially greater impact on detection performance than random features.

Additionally, analyzing the feature ranking sorted by average importance indicates that *FIRST_SWITCHED*, *FLOW_DURATION_MICROSECONDS*, *L4_DST_PORT*, *L4_SRC_PORT*, and *FLOW_ID* stand out as the most influential in classifying anomalies. By prioritizing these high-impact features, analysts can concentrate on the areas most likely to yield actionable insights, thereby reducing cognitive load and streamlining the supervisory process.

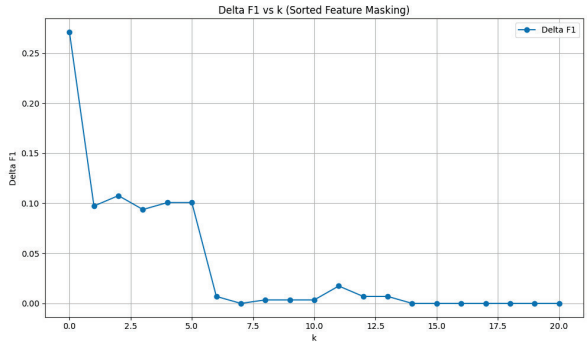

Fig. 6. Net change in F1 (xAAD vs. baseline) when removing top-k xAAD features.

C. Sparsity (Metric 2)

Measures the degree to which a small subset of features accounts for most of the detected anomalies. In Figure 8, we see that many features exhibit near-zero importance scores, implying that xAAD's anomaly detection relies heavily on only a few discriminative features.

From an operational standpoint, this sparsity is a significant advantage. Security operations centers (SOCs) frequently deal with high-dimensional data and must sift through numerous attributes (IP addresses, ports, protocols, timing intervals, etc.) to find relevant leads. If an explainable model shows that only a handful of features drive most anomaly decisions, it not only saves analysts time but also helps them identify consistent attack vectors or vulnerabilities. By focusing on the most salient features, organizations can streamline threat-hunting efforts and refine monitoring rules without sacrificing detection coverage.

D. Robustness (Metric 3)

We evaluated by perturbing random subsets of features with Gaussian noise proportional to each feature's standard deviation. Figure 9 presents a heatmap of the correlations

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between the original and perturbed explanations under increasing noise levels and fractions of disturbed features. Even at moderate noise levels, the correlation remains high, suggesting that xAAD's feature importances are stable against minor input fluctuations.

In practical terms, this robustness translates to greater reliability when network conditions change or benign anomalies arise (e.g., sudden but legitimate traffic surges). If slight deviations in network flow metrics do not substantially alter xAAD's explanation of why a flow is deemed anomalous, then security teams can trust the model's outputs across varying real-world conditions. This consistency is key to ensuring that the explainability provided by xAAD remains valuable when scaling up to more volatile network environments.

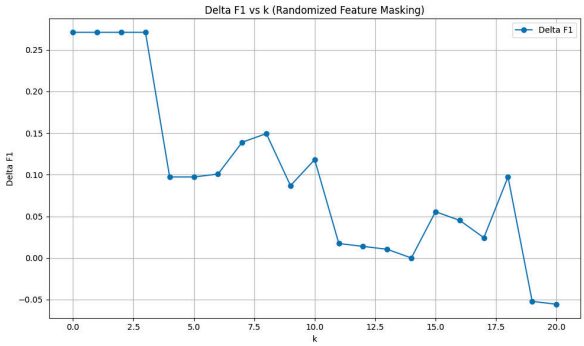

Fig. 7. Net change in F1 when removing *random* features, showing a gentler performance impact.

E. Discussion of Overall Findings

Collectively, these results confirm that xAAD brings tangible benefits to cybersecurity anomaly detection:

- *Improved Detection Focus:* The t-SNE plots illustrate that iterative analyst feedback sharpens the boundary between normal and abnormal traffic, pointing to fewer false positives and more reliable anomaly groupings.
- *Meaningful Explanations:* Descriptive Accuracy tests indicate that xAAD identifies features truly driving anomalies. This granularity reduces the guesswork analysts face when investigating alerts.
- *Efficient Feature Utilization:* The strong Sparsity of xAAD's explanations allows security personnel to concentrate on critical traffic attributes rather than examining all 44 original features.
- *Stable Insights:* Robustness against noise ensures that explanations remain consistent even when network conditions or measurement precision vary, reducing the chance of explanation drift in production.

In high-stakes domains like cybersecurity, such properties are invaluable. By aligning technical detection outputs with actionable and explainable insights, xAAD addresses the longstanding gap between purely algorithmic anomaly detection and the human-centric processes of incident response, threat mitigation, and post-event analysis.

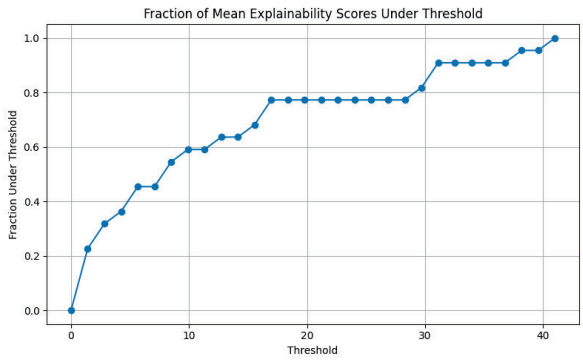

Fig. 8. Sparsity plot: Fraction of features below each importance threshold for SIMARGL2021.

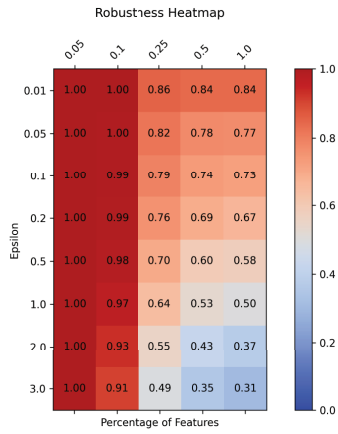

Fig. 9. Robustness heatmap: correlation of original vs. perturbed xAAD explanations at varying noise levels and fractions of perturbed features.

VI. Conclusion and Future Work

We applied xAAD to the RoEduNet-SIMARGL2021 Network Intrusion Detection Dataset and evaluated it using three key E-XAI metrics: Descriptive Accuracy, Sparsity, and Robustness [9]. Our findings show that xAAD effectively isolates the most critical features underlying anomalous network behaviors, emphasizing a smaller subset of highly informative attributes. This not only improves detection performance but also offers transparent, actionable insights for security analysts seeking to understand *why* certain traffic flows are classified as suspicious. In addition, xAAD's resilience to perturbations suggests that its explanations maintain consistency even amid fluctuating network conditions, a valuable property in volatile cybersecurity environments.

Looking ahead, we plan to further expand xAAD's capabilities by incorporating additional E-XAI metrics such as stability and completeness, enabling a more thorough assessment of explanation quality. Another direction of interest is to exploring how xAAD can be adapted for continuous, real-time data streams, allowing on-the-fly anomaly detection and explanation. By pursuing these directions, xAAD can evolve into a more comprehensive and adaptable framework, meeting the increasing demands

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for transparent and efficient anomaly detection in high-stakes cybersecurity settings.

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